

Pappu Jha

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Education

The University of Southern Mississippi (USM)	Hattiesburg, Mississippi
Bachelor of Science (BS) in Computer Science	GPA: 4.0/4.0
Minors: Economic Data Analysis, Information Security	Graduation: May 2027
Honors Keystone Scholar, TEDx Southern Miss 2025 Speaker	
Coursework: <u>Designing Solutions for Defense</u> , Data Structures & Algorithms, Econometrics, Machine Learning	

Work Experiences

Data Analytics & Business Intelligence Intern (Aviation) , The Port Authority of NY & NJ – New York, NY	May 2026 – August 2026
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- Developing an **ML model** in **Python** on **Databricks** to forecast hourly passenger volumes at **Newark Airport (EWR) Terminals A & B** using indirect datasets, bridging PANYNJ's lack of access to airline-held passenger data.
- Building **Power BI dashboards** to translate hourly demand forecasts into actionable insights, enabling data-driven staffing and resource allocation at one of the nation's busiest airports (**47M+ passengers in 2025**).

IT Data Analyst Intern , Jones Capital – Hattiesburg, MS	Sept 2025 – May 2026
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- Designed and modified 10+ automation workflows in **Jira Service Management (JSM)** across over 8 portfolio companies, reducing service request resolution time by 25% and aligning delivery with **SLA** targets.
- Built 30+ dynamic gadgets in JSM and **Excel** using **JQL filters** to monitor real-time performance of 5 IT agents, resulting in a 30% increase in team efficiency and improved KPI tracking.
- Developed 5 **automation pipelines** in **n8n**, created 20+ pages of process **documentation** in **Confluence**, and published 5 **Scribe** tutorials, reducing manual reporting time by 40%.
- Resolved 10+ field-level discrepancies in more than **2000 JSM work items** of all time by troubleshooting, leading to a **100%** reduction in recurring ticket errors in the following weeks.

Data Engineer/Research Assistant , Institute for Advanced Analytics and Society – Hattiesburg, MS	May 2025 – May 2026
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- Built **AQUAVIEW**, a Python library exposing more than 7 API endpoints for streamlined access to oceanographic data for NOAA (National Oceanic and Atmospheric Administration), downloaded over **1000 times**.
- Processed **17M+ data** from **125 years** of the **World Ocean Database**, generated metadata in MongoDB, indexed in Cloud, and integrated into the UI via Elastic Search, contributing to the NOAA's AQUAVIEW project.
- Enhanced the user interface of World Ocean Database to handle **edge cases**, including invalid date ranges and out-of-bound latitude/longitude values, improving overall user experience.
- Developed an **LLM-powered classification tool** to assign Platform IDs for Uncrewed Marine Systems using metadata in real-time, streamlining NOAA's platform designation across **20M+ data points** in **10+ sources**.
- Created an adapter for the **National Data Buoy Center (NDBC)** historical and real-time data from inactive and active stations, enabling easy access to over **30 years** of NDBC datasets.

Web Developer , Roy Howard Community Journalism Center – Hattiesburg, MS	Oct 2024 – May 2025
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- Developed and maintained the organization's content-heavy website using **WordPress (SNO CMS)**, improving user engagement by **35%** through targeted UI/UX redesign and responsive layouts.
- Enhanced visibility by optimizing **SEO metadata**, improving page load speed by **40%**, and indexing high-traffic pages on **Google Search Console**.
- Collaborated with editorial and tech teams to ensure site performance, troubleshoot errors, and publish timely content updates, ensuring **99.9% uptime**.

Research Experience

Undergraduate Research Assistant (Volunteer), Cyber Innovation Lab **Sept 2024 – Present**

- Led two cybersecurity research projects using machine learning and deep learning in collaboration with faculty and mentors, resulting in a conference presentation, research award, and publication.

Creative Projects

Boston Daddy [Link](#)
Hack Harvard 2025, Harvard University **Oct 2025**

- Developed a **full-stack** smart city infrastructure platform that integrated energy, traffic, and weather data — achieving predictive analytics and actionable insights for 15+ buildings and 10+ intersections in real-time.

Delagent [Link](#)
MHacks 2025, University of Michigan **Sept 2025**

- Built an AI-based scheduling platform using **Fetch.AI uAgents framework, FastAPI, and SQLite**, resulting in automated 3-agent calendar coordination and negotiation with 80% reduction in manual scheduling overhead.

Regulating College Dormitories Digitally Using C++ [Link](#)
Advisor: Mr. Rick Loggins, USM Faculty Member **Nov 2023**

- Built a dormitory management back-end system using all concepts from the introductory C++ course, streamlining student check-ins/outs and visitor logs, reducing manual tracking effort by 60%.
- Integrated rule enforcement features to automatically flag dorm violations, improving housing compliance oversight of 500+ students.

Publication & Presentation

Conference Proceeding

- **Jha, P.**, Hamid, H., Olukola, O., Dahal, A., & Rahimi, N. (2026). *Adversarial Machine Learning for Robust Password Strength Estimation*. In: Rahimi, N., Margapuri, V., & Golilarz, N.A. (Eds.) *Software and Data Engineering. SEDE 2025*. Communications in Computer and Information Science, vol. 2720. Springer, Cham. https://doi.org/10.1007/978-3-032-08649-5_18
- Yamali, S., **Jha, P.**, Kandel, B., Poudel, A., & Hill, J. (2026). World Ocean Database: Building a search engine for 200 years of ocean data. Oral presentation and poster presented at the UG2 (Underwater Glider User Group) Workshop 2026, St. Petersburg, FL.

Presentations

- USM Undergraduate Symposium on Research & Creative Activity – **First Place** (Computational Approaches category), **Adversarial Machine Learning on Robust Password Strength Estimation** (Apr 2025)
- Golden Idea Pitch Competition, **Top 8 Finalist** – USM Center for Entrepreneurship (May 2026)
- USM Undergraduate Symposium on Research & Creative Activity, **Social Engineering Attack Detection in Corporate Communications** (May 2026)

Skills

Programming: Python, C++, OOP, API, SQL, Web Development (HTML, CSS, JavaScript, WordPress)

Information Technology: Atlassian (Jira Service Management, Confluence), n8n, Power BI, Scribe, Microsoft 365 tools, Cloud Services, GitHub, Database (MongoDB), SharePoint, OneDrive, Jupyter Notebook, LLM/GenAI

Data Analytics: Power BI, Python (NumPy, Pandas, Scikit-learn, Seaborn), SQL, Machine Learning Classification & Regression (Random Forest, Logistic Regression, Naive Bayes), Object Oriented Programming (OOP)

Leadership Experiences

Creative Director, The Entrepreneurship Society, USM Center for Entrepreneurship **Aug 2025 – Present**

- Conducted marketing of the organization via social media, mass emailing, and distributing flyers, increasing the membership to more than 80 in a semester.

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| International Ambassador/Mentor , USM Center for International Education | Dec 2023 – Present |
| <ul style="list-style-type: none"> Supported 100+ international students by managing pre-arrival communication on Unibuddy and on-site onboarding to ensure a smooth transition to the United States. | |
| Director of Finance , USM Residence Hall Association | Apr 2024 – May 2025 |
| <ul style="list-style-type: none"> Formulated budgets for 8+ RHA events with 1000+ attendees, coordinating with board members and advisors to ensure successful execution and efficient resource allocation. | |

Honors & Awards

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| USM President's Lists : Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026 | Aug 2023 – May 2026 |
| First Place , Undergraduate Symposium for Research and Creative Activity (UGS) | Apr 2025 |
| International Ambassador of the Year , USM Center for International Education | Apr 2025 |
| Innovation & Excellence Award , Roy Howard Community Journalism Center, USM | Apr 2025 |
| Honors Keystone Scholarship , USM Honors College | Aug 2025 - May 2027 |
| Academic Excellence Scholarship , The University of Southern Mississippi | Aug 2023 - May 2027 |
| Top 500 Global Finalist , Rise For The World | Jan 2022 |